

VANZHILKYNAK, Russian Federation

WMO: 239660

Lat: 60.350N

Lon: 84.083E

Elev: 99

StdP: 100.14

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.1	-39.9	-44.8	0.0	-42.7	-41.8	0.1	-39.6	4.5	-12.1	4.1	-13.1	0.1	300	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.7	19.3	26.9	18.5	24.9	17.6	20.6	26.5	19.4	24.9	18.4	23.4	2.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	23.1	17.5	12.6	21.9	16.4	11.8	20.8	60.1	26.5	56.1	25.1	52.6	23.5	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.5	3.9	3.3	DB	-44.9	31.3	3.8	1.4	-47.7	32.4	-49.9	33.2	-52.1	34.0	-54.9	35.0	
			WB	-44.5	22.6	3.4	1.5	-47.0	23.7	-49.0	24.5	-50.9	25.4	-53.4	26.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.2	-22.6	-18.5	-9.0	-1.5	6.1	14.8	17.3	13.7	7.1	-1.0	-13.9	-19.8
	DBStd	15.42	9.53	8.48	7.31	6.12	5.76	5.06	3.68	3.82	4.48	5.78	9.95	10.27
	HDD10.0	5014	1012	799	589	346	151	16	1	10	109	342	717	922
	HDD18.3	7551	1271	1032	848	596	379	126	64	151	337	600	967	1181
	CDD10.0	565	0	0	0	0	30	160	227	124	23	1	0	0
	CDD18.3	61	0	0	0	0	1	21	32	7	1	0	0	0
	CDH23.3	742	0	0	0	0	34	265	362	77	4	0	0	0
	CDH26.7	172	0	0	0	0	7	66	86	13	0	0	0	0
	WSAvg	1.2	0.9	1.0	1.3	1.4	1.5	1.2	1.1	1.1	1.2	1.3	1.2	1.0
Precipitation	PrecAvg	626	30	26	29	34	52	77	83	76	67	59	51	41
	PrecMax	783	65	61	68	64	115	153	187	155	136	107	83	79
	PrecMin	408	2	8	6	3	21	35	10	2	19	17	22	12
	PrecStd	105	15	11	15	16	23	28	46	38	27	19	18	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	0.0	7.6	16.0	27.3	30.8	31.3	28.7	23.4	13.9	3.1	-0.3
		MCWB	-1.5	-1.0	3.4	8.8	17.5	20.0	21.3	20.2	16.1	10.0	2.1	-0.8
	2%	DB	-4.0	-2.3	4.7	12.8	22.9	28.7	28.9	25.5	19.6	10.7	1.3	-2.1
		MCWB	-4.6	-3.2	1.0	7.5	14.2	18.8	19.9	18.2	14.2	7.8	0.8	-2.6
	5%	DB	-6.6	-4.6	2.5	10.2	19.4	26.7	27.2	23.1	16.8	8.5	0.5	-3.7
		MCWB	-7.0	-5.4	0.1	5.6	12.3	17.7	19.1	17.0	12.9	6.4	0.1	-4.2
	10%	DB	-9.2	-7.0	0.7	7.8	16.0	24.3	25.3	20.9	14.3	6.2	-0.7	-5.9
		MCWB	-9.6	-7.8	-1.0	4.0	10.5	16.8	18.2	16.1	11.5	4.9	-1.2	-6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-0.7	3.8	10.3	17.7	21.3	23.0	20.9	16.8	10.3	2.3	-0.7
		MCDB	-1.3	-0.2	7.4	15.1	26.2	27.8	29.8	27.2	22.0	13.0	2.8	-0.4
	2%	WB	-4.5	-3.1	1.7	7.7	14.9	19.7	21.2	19.2	14.8	8.3	0.9	-2.5
		MCDB	-4.0	-2.5	3.8	12.0	21.6	26.3	27.0	24.0	18.4	10.3	1.1	-2.1
	5%	WB	-7.0	-5.4	0.3	6.0	12.9	18.6	20.0	17.9	13.2	6.6	0.2	-4.2
		MCDB	-6.6	-4.7	2.2	9.9	18.3	24.9	25.2	21.6	16.3	8.2	0.6	-3.7
	10%	WB	-9.7	-7.7	-1.0	4.1	10.9	17.6	19.0	16.6	11.7	4.9	-1.2	-6.4
		MCDB	-9.2	-7.0	0.8	7.3	15.6	23.1	23.9	20.3	14.1	6.2	-0.7	-5.9
Mean Daily Temperature Range	5% DB	MDBR	8.4	10.2	11.8	11.2	10.6	11.2	10.9	10.0	8.4	6.3	7.5	7.6
		MCDBR	9.1	8.7	12.3	13.9	16.3	15.4	14.4	13.9	13.5	9.3	5.0	7.9
		MCWBR	8.9	8.2	9.5	9.4	9.1	6.8	6.6	7.5	8.3	6.6	5.1	7.9
	5% WB	MCDBR	9.2	8.8	11.0	13.0	14.8	13.3	12.5	11.9	12.1	8.4	5.2	8.0
		MCWBR	9.1	8.3	8.8	9.1	8.7	6.6	6.5	7.1	7.7	6.2	5.3	8.0
Clear-Sky Solar Irradiance	taub		0.265	0.301	0.347	0.369	0.383	0.400	0.443	0.410	0.359	0.326	0.270	0.255
	taud		1.961	2.006	1.964	2.046	2.275	2.329	2.184	2.264	2.372	2.257	2.016	1.904
	Ebn at Noon		445	655	745	814	840	830	775	773	757	640	435	289
	Edh at Noon		49	84	123	137	118	114	129	110	83	66	45	33
All-Sky Solar Radiation	RadAvg		0.45	1.36	2.84	4.41	4.86	5.56	5.03	3.68	2.23	1.03	0.47	0.23
	RadStd		0.05	0.11	0.19	0.44	0.61	0.52	0.53	0.43	0.30	0.11	0.10	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page